pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

pandas: Combining and Loading, Part 1

WEEK 2, DAY 3, PART 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

By the end of this part you can

Join tables on a shared column

The job VLOOKUP does, with the row count checked.

```
pd.merge(holdings, ratings, on='rating')
```

Choose how to join

Inner, left, right, and outer, and what did not match.

```
pd.merge(a, b, on='ticker', how='left')
```

Stack tables

Add rows, and check that nothing was lost.

```
pd.concat([ig_book, hy_book])
```

Summarize by group

Counts, totals, and averages. In Excel, a pivot table.

```
book.groupby('bucket')['oas'].mean()
```

Yesterday one table. Today, several.

Each file comes from a different system. A shared column, the key, links one to the next.

`positions.csv`

200 bonds

Six columns. A ticker, and nothing else about the issuer.

`ticker` →

`issuers.csv`

150 issuers

Name, sector, country, and rating. One row per ticker.

`rating` →

`ratings.csv`

18 ratings

A score, a bucket, and a grade for each rating.

`holdings.csv`

200 bonds

Yesterday's file. The same bonds, with all 19 columns.

`cusip` →

`benchmark.csv`

821 bonds

What the portfolio is measured against. Invented, too.

Two ways to combine

`merge`
side by side, adds columns

`concat`
stacked, adds rows

merge: a lookup across tables

```
named = pd.merge(positions, issuer_details, on='ticker')
```

on= names the column to match on: the key. It appears once in the result.

positions: three of its six columns

ticker	coupon	par_held
BEZE	7.000	5000000
BEZH	8.125	250000
BEZL	5.125	1500000

+

issuer_details: one row per ticker

ticker	issuer	sector	rating
BEZE	Bezane Finance	Financials	BBB-
BEZH	Bezamyth Machinery	Industrials	B
BEZL	Bezoquil Freight	Transportation	BB

Check the count

200 rows in, 200 rows out. Six columns became nine.

named: every position now carries its issuer, its sector, and its rating

ticker	coupon	par_held	issuer	sector	rating
BEZE	7.000	5000000	Bezane Finance	Financials	BBB-
BEZH	8.125	250000	Bezamyth Machinery	Industrials	B
BEZL	5.125	1500000	Bezoquil Freight	Transportation	BB

In Excel

```
=VLOOKUP(B2, issuers!A:C, 3, FALSE)
```

Filled down 200 rows, once for each column wanted.

how= decides which rows are kept

desk_small

ticker	face
QVNE	1500000
DNMR	500000
PLWK	750000
HLVF	250000

sector_small

ticker	sector
QVNE	Energy
DNMR	Transportation
PLWK	Materials
THAR	Materials

```
left_join = pd.merge(
    desk_small, sector_small,
    on='ticker', how='left')
  ticker     face          sector
0   QVNE  1500000          Energy
1   DNMR   500000  Transportation
2   PLWK   750000       Materials
3   HLVF   250000             NaN
```

how='inner'

Keys found in both tables. The default.

QVNE DNMR PLWK ~~HLVF~~ ~~THAR~~

3 rows HLVF and THAR dropped

how='left'

Every row of the left table. Like a VLOOKUP.

QVNE DNMR PLWK HLVF ~~THAR~~

4 rows HLVF kept, its sector is NaN

how='right'

Every row of the right table.

QVNE DNMR PLWK ~~HLVF~~ THAR

4 rows THAR kept, its face is NaN

how='outer'

Every key from either table.

QVNE DNMR PLWK HLVF THAR

5 rows Nothing dropped

A merge can add rows, and no error says so

desk_small: 4 rows

ticker	face
QVNE	1500000
DNMR	500000
PLWK	750000
HLVF	250000

+

sector_twice: QVNE is listed twice

ticker	sector
QVNE	Energy
QVNE	Energy
DNMR	Transportation
PLWK	Materials
HLVF	Consumer

→

doubled: 5 rows from a left merge

ticker	face	sector
QVNE	1500000	Energy
QVNE	1500000	Energy
DNMR	500000	Transportation
PLWK	750000	Materials
HLVF	250000	Consumer

Face before the merge

3,000,000

4 rows in the desk's table.

→

Face after the merge

4,500,000

5 rows. QVNE is counted twice.

The habit

Print the row count before and after every merge.

The cure: a lookup table with each key once. Check it with nunique().

A guard written into the code

A repeated key: validate= stops the merge

```
pd.merge(desk_small, sector_twice,
         on='ticker', how='left',
         validate='many_to_one')
MergeError: Merge keys are not unique in
right dataset; not a many-to-one merge
```

many_to_one: many positions on the left, one row per key on the right. No table comes back, so nothing is doubled.

In Excel VLOOKUP returns the first match and gives no sign of a second. The test: `=COUNTIF(A:A, A2)` on the lookup sheet, filtered for values above 1.

On the real lookup it passes, and an assert checks the row count

```
guarded = pd.merge(positions, issuer_details, on='ticker', how='left', validate='many_to_one')
assert guarded.shape[0] == positions.shape[0], 'the merge changed the row count'
Rows in: 200  rows out: 200
```

The portfolio against the benchmark, as counts

```
desk_face = holdings[['cusip', 'par_held']]
compared = pd.merge(benchmark, desk_face,
                    on='cusip', how='left')
print('Rows:', compared.shape[0])
Rows: 821
```

cusip	ticker	weight_pct	par_held
99001BAA4	QVNE	0.2076	2750000.0
99001BAB2	QVNE	0.1551	NaN
99001BAC0	QVNE	0.0245	NaN
99001BAD8	QVNE	0.2024	NaN

The benchmark's 821 bonds. Each box is as wide as its share of the benchmark's weight.

200 held
24.56 percent of the weight

621 not held
75.44 percent of the weight

`left_only`

621 In the benchmark alone

`both`

200 In both files

`right_only`

0 Held, and not in the benchmark

Two invented files. Counts and weights that describe the overlap. Not a view.

concat stacks tables and adds rows

Two tables with the same 22 columns

ig_book
128 rows

hy_book
72 rows

→

stacked
200 rows, 22 columns
Market value 490,819,215

```
stacked = pd.concat([ig_book, hy_book])
print('Stacked:', stacked.shape)
Stacked: (200, 22)
```

Row labels where the first table ends and the second begins

As stacked	198	199	**1**	**2**
`ignore_index=True`	126	127	128	129

merge

Side by side. Adds columns. Matches the rows on a key.

concat

Stacked. Adds rows. Lines the columns up by name.

In Excel

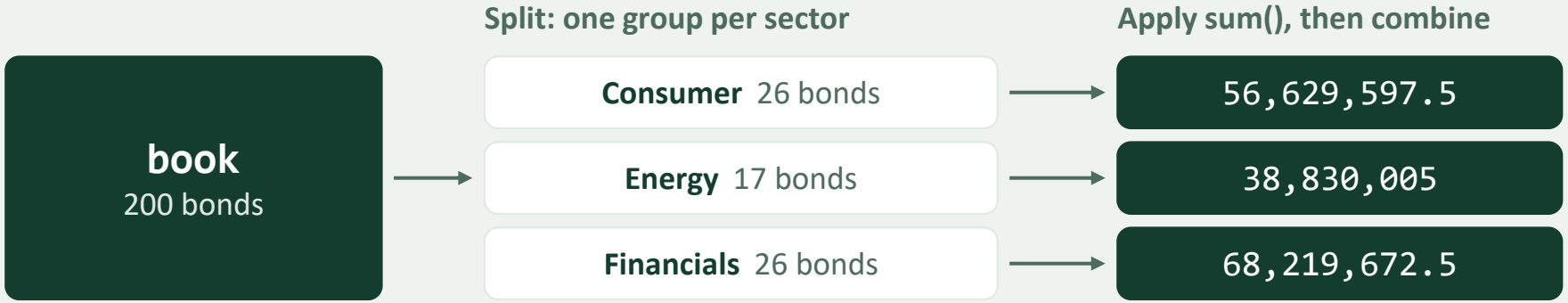
Paste one sheet under another, or =VSTACK(). Excel pastes by position.

groupby: split, apply, combine

```
book.groupby('sector')   ['market_value']   .sum()
```

How to split the rows | The column to use | The calculation

In Excel, a pivot table: sector in Rows, Sum of market_value in Values.



Seven more sectors follow. Ten totals, in alphabetical order. The index is the sector.

The same pattern	pandas
A mean, by rating bucket	`book.groupby('bucket')['oas'].mean()`
Several statistics at once	`book.groupby('grade')['oas'].agg(['count', 'mean', 'median', 'min', 'max'])`
A summary table, column by column	`book.groupby('sector').agg(bonds=('cusip', 'count'), ...)`

The AI check

Every notebook ends with code written in the style of an AI assistant. It has one bug. Find it.

1

Read

Read the code before you run it.

2

Predict

Decide what it should produce.

3

Run

Run it and look at the output.

4

Check

Compare with a number you worked out yourself.

Code that runs is not the same as code that is right.

What you can do now

- Join tables, and check the row count every time
- Choose how to join, and guard the merge with validate=
- Compare a portfolio with its benchmark by CUSIP
- Stack tables, and put them side by side
- Summarize by sector, rating bucket, or grade

Next pandas: Combining and Loading, Part 2: weighted averages, saving results, dates, and workbooks